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METHODOLOGIES AND APPLICATION

Pre-production box-office success quotient forecasting

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Abstract

Hollywood enjoys the position of being the biggest movie producers when it comes to global recognition among movie-making industries. Despite being the biggest movie producer, it has been facing high revenue losses lately since most of the films that it has created have failed to capture viewer's attention in the first few weeks of its release resulting in a box-office flop. It has been observed in a recent study that Hollywood is estimated to witness a loss of around 1 billion to almost 10 billion US dollars till 2020. Revenue risks have created immense pressure on movie producing stakeholders. They feel a constant pressure to come up with a formula to make a successful movie, however, to date; there are no fixed ingredients that can ensure the success of a movie. Researchers and movie producers constantly feel a need to have some expert systems which would predict the fate of the movie prior to its production with reasonable accuracy. Regardless of the difficult nature of the issue area, few researchers have created expert systems to forecast the financial success of movies using different approaches, but most of them are targeted pre-release forecasting or have low prediction accuracy. Such predictions are of a seminal nature as of their limited prediction scope, and non-ability to reduce revenue loss risk. Therefore, there is a constant demand from investors to have pre-production forecasting tools with high accuracy which can help them plan and make necessary alterations to save huge investments. In this study, we proposed eighteen new features to forecast box-office success, as soon as the quotient (director and cast) signs an agreement. This proposed forecasting time is the earliest prediction that has ever been reported in the movie forecasting literature. The decision support system ranks director and lead cast by utilizing their performances of the last 100 years (1915–2015). The processed output file is a table that ranks each director and cast into four categories based on cast experience, journalist critics, media reporting, user ratings, and revenue generated by associating movie. To produce more accurate results, learner-based feature selection is also performed to select the best subsets of features. This system is intended to be a dynamic tool, integrating further data for real-time adaptation. The system has the ability to incorporate different feature selection algorithm for the progressive improvement of movie success forecasting We demonstrate the effectiveness of extracting features and explain how they improve forecasting accuracy over existing models. The adaptive behaviour of the presented system is achieved by incorporating conceptually different machine learning classifiers, i.e. support vector machine, gradient boosting, extreme boosting classifier, and random forest. A voting system is used to make the prediction by averaging the output class-probabilities. To assess the adequacy of new features, a cross-validation test is directed. Our classification results are evaluated by using two performance measures, i.e. average per cent success rate, or within one class away from its actual prediction. The new features have achieved the most noteworthy accuracy of 85% with an expansion of a 46.43% (average per cent success rate) and 5.56% (within a class away) in comparison with other state-of-the-art feature sets.

Keywords Machine learning · Classification · Hollywood · Facebook · Forecasting

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1 Introduction

Hollywood releases around a hundred movies every year with anticipation to reciprocate investment that would only be possible if a movie succeeds. For instance, people have a variety of interests when it comes to movies. Some watch comedy, some like romance, and others would prefer science fiction or thriller. There are some who would watch a movie solely because of a particular actor or actress has worked in it. A common challenge encountered by all movie producers is that they are expected to cater to the desires of a huge mass of audience regardless of the fact that likes, and dislikes differ subjectively. To keep a diversified audience entertained and anticipated is an excruciating problem, as entertainment is something which cannot be quantified at all. Hollywood is always expected to amuse the audience and to achieve massive success. However, the question that arises is how to predict the degree of entertainment offered by a certain movie or the other way to its success or failure? Is there any way to predict how successful a movie is going to be before it's all set to release or even before its production starts? Let me quote, Jack Valente, President and CEO of (MPAA) who once said, "No one can tell you how a movie is going to do in the marketplace. Not until the film open in darkened theatre and sparks fly up between the screen and audience". This statement enlightens us about the level of complexity involved in predicting the success of a movie. It is interesting to note that success, here, is termed as financial success at the box office.

It is evident that in today's market every innovation or the new product being released is finally facing the ever-changing judgment of people which also gives an insight into the interests of the public. However, one can narrow the outcome of uncertainty by using the product historical data which contains different hidden patterns. In the year 2016, the Hollywood industry was expected to produce the blockbuster, but only a few movies fell into the blockbuster category. The vast amount of Hollywood movies released during summer break in 2016, generated about 40 per cent of annual sales. In 2015 Hollywood's market results revealed that industry had shifted its focus from strategy to making costly remakes and sequels. The year 2016, which was the industry's peak season, appeared as a disappointment even though ticket sales were about equal to 2015's haul (Nash Information Services, LLC). Out of 32 summer season movies released, 19 by the six major studios, 17 lost a total of US dollar 915.6 million (The Numbers). Last year the studios released a total of 15 box-office flops with loss of US dollar 546.3 million. Hence, the statistics show that the amount of loss is prone to increase in the coming years.

This study analyses the movie's database and social media's content to predict box-office success at the pre-production level. Furthermore, it has also been taken into account how these features can be combined with future research work. The research explores Hollywood data for the last 100 years (1915–2015) to predict the success (profit) at the box office of movies in 2016. This paper has presented eighteen new features to classify each movie as the blockbuster or a flop. The trained model can make a prediction at an early stage of the film release. The prediction accuracy is high as 85%, which can provide verdict reference for investors and reduces the risk associated with that verdict. A diverse examination is performed to check the impact of these features in anticipating the success of the motion picture.

Accurate forecasting prior to the film release has a very practical implementation in risk control and decision-making. This helps the production house to select and build a right combination of the cast that would have a great reputation within the industry and have an influential impact on the audience. This study will help all the stakeholders associated with Hollywood to foresee the chances of success for a particular movie provided those explanatory variables before signing any project.

1.1 US movie industry 1995–2016

Currently, the US box office is occupied by 10 major studios. They shared a market of over 94.20% as mentioned in Table 1. The studio's functionality generally depends upon the finance, production, distribution, and advertisement (Vogel 2001). In order to be not involved in the controversy regarding different obligations, theatre chains and individuals separate themselves from the studios (Eliashberg et al. 2000, 2006).

Revenue generated from ticket sales is very broad in nature. During the past two decades, billions of tickets have been sold with an average of $11 billion as mentioned in Table 2. Every year there is the expansion of almost 32% (Numbers 2016). The nature of the market is very erratic since it is not easy to predict the consequences beforehand. It can either be a hit or a complete flop. The income generated by blockbuster has kept on increasing Eliashberg et al. (2006). The cost of filmmaking is expanding from under 30 million (the 1990s) to 65 million (mid-2000s) Eliashberg et al. (2006) (Table 3).

During the period of 2000–2010, more than 75 films were subjected to a deficiency of $50 million in comparison to the production budget with domestic industry income. Out of 3453 movies, only 31% lost 16.5 million of their budget and were not able to recover their investment (Table 2) (Ghiassi et al. 2015).

1.2 Box-office art of revenue generation

The movie distribution is more difficult than movie production. Because of this fact, the studio makes an agreement with well-reputed movie distributors to generate good revenue. Different distributors secure the deal by having a back or relation to the director or cast. Production houses usually use

Table 1 Movie distributor ranking

Rank	Distributor	Share in percentage	Revenue US dollars	Number of movies
1	Buena Vista	26.40	3000.90	16
2	Warner Bros	16.80	1907.60	37
3	20th Century Fox	12.90	1469.10	23
4	Universal	12.30	1405.50	20
5	Sony/Columbia	8.00	911.20	26
6	Paramount	7.70	876.80	18
7	Lionsgate	5.80	665.00	27
8	Focus Features	1.70	196.40	12
9	STX Entertainment	1.70	195.60	7
10	Open Road Films	0.90	107.00	8
	Total average	94.20	10, 735.10	194

Table 2 Aggregate loss (2000–2010) mentioned in Ghiassi et al. (2015)

Total films released	2079 films
Average production budget $ USD	$35,737,733
Average production budget $ USD	$35,737,733
Average US box-office revenue $ USD	$44,213,244
Average worldwide box-office revenue $ USD	$111,986,081
Gross domestic box-office receipts > production budget	968 films
Average profit $ USD	$37,235,560
Gross domestic box-office receipts < production budget	1111 films
Average domestic loss ($ USD)	$16,582,750
Gross global box-office receipts < production budget	445 films
Average global loss $ USD	$12,839,501

Table 3 Average ticket price (information mentioned by an online movie database site (The Numbers n.d.)

Year	Tickets sold	Total box office	Average ticket price
2016	1,334,591,023	$11,250,605,727	$9.80
2015	1,340,684,258	$11,301,971,695	$8.43
2014	1,267,678,776	$10,356,938,680	$8.17
2013	1,340,634,466	$10,899,361,167	$8.13
2012	1,384,375,983	$11,019,635,771	$7.96
2011	1,284,682,574	$10,187,535,273	$7.93
2010	1,331,635,139	$10,506,603,479	$7.89
2009	1,418,841,209	$10,641,310,931	$7.50
2008	1,362,438,798	$9,782,312,632	$7.18
2007	1,419,657,445	$9,767,245,765	$6.88
2006	1,403,069,991	$9,190,110,914	$6.55
2005	1,376,193,127	$8,821,400,084	$6.41
2004	1,494,942,916	$9,283,597,458	$6.21
2003	1,524,980,632	$9,195,635,095	$6.03
2002	1,577,407,684	$9,164,740,433	$5.81
2001	1,465,427,360	$8,294,320,508	$5.66
2000	1,398,001,724	$7,535,230,958	$5.39
1999	1,444,717,431	$7,339,165,852	$5.08
1998	1,443,007,684	$6,767,707,031	$4.69
1997	1,385,216,757	$6,358,145,799	$4.59
1996	1,310,013,419	$5,790,260,095	$4.42
1995	1,221,825,463	$5,314,941,459	$4.35

film festivals to get the attention of distributors. After that, both parties make proposals to secure better deals. Meanwhile, in the box office, two models are in the process, i.e. leasing and profit sharing. In the leasing model, the distributor pays a fixed amount for the movie rights. On the other hand, the distribution and production house get a percentage (10–50%) on the net profit of the movie. Both models are dependent on the performance of the movie at the box office. This research will benefit both the distributor and the production house in order to make a verdict. Top grossing distribution is mentioned in Table 4.

Some researchers have tried utilizing the initial theatre performance of a Hollywood movie to forecast the success. In cinemas, the first week of the performance of a movie is very critical as it must generate significant income within a limited time. Media reporter and critics create the buzz in the audience; resultantly more audience visits the cinema (Vogel 2001; Elberse and Eliashberg 2003). Buzz created by the critics ignites the demands of the public which help the film run in cinemas for an extended period. The distributor does not only take the rights of domestic and international release but also of home video and network television release.

Table 4 Average ticket price information mentioned by an online movie database site (The Numbers n.d.)

Rank	Distributor	Movies	Total gross	Average gross	Market share (%)
1	Walt Disney	531	$30,343,472,350	$57,144,016	15.24
2	Warner Bros.	678	$29,978,549,069	$44,216,149	15.06
3	Sony Pictures	642	$24,663,402,316	$38,416,515	12.39
4	20th Century Fox	470	$23,104,581,842	$49,158,685	11.61
5	Universal	434	$22,571,427,351	$52,007,897	11.34
6	Paramount Pictures	441	$22,238,589,087	$50,427,640	11.17
7	Lionsgate	343	$7,405,311,611	$21,589,830	3.72
8	New Line	205	$6,193,114,702	$30,210,316	3.11
9	Dreamworks SKG	78	$4,278,670,768	$54,854,753	2.15
10	Miramax	382	$3,841,082,392	$10,055,190	1.93

By choosing the right strategy, both production house and distributor try to generate the maximum revenue out of the movie (Vogel 2001; Elberse and Eliashberg 2003).

The foundation for this exploration is the data sets we have developed to some extent by harvesting a scope of freely accessible film information. Therefore, our data sets are more widespread in nature than those already referred to in the literature. Our research focuses on improving the accuracy of pre-production revenue forecast model proposed by Ghiassi et al. (2015), Sharda and Delen (2006), Delen and Sharda (2010) by using new proposed features. The goal of this study is to develop a forecasting model that would make prediction easier because of its practical value for investors or verdict maker in the film industry.

This paper is organized as follows. In Sect. 2, the overview of existing state-of-the-art approaches to movie forecasting is presented. Section 2.6 provides details of the proposed methodology, by enlightening the new data acquisition, proposed features, machine model, and performance measure experiment results. Next, forecasting result and comprehensive comparison with the different model are explained in Sect. 3. Lastly, Sect. 4 of the paper describes the contribution of this study and further research direction.

2 Literature review

Producing a motion picture has always been expensive and risky venture due to its unpredictable consequences. Production studios normally do not employ forecasting models or computer algorithms, to predict a movie's success. To mitigate the possibility of loss, studies have found a solution, i.e. to maximize a budget for a movie. It does not stay there and become a risky endeavour when studios try to pair star power and expensive budget since they consider it as a guarantee for success. Since a huge amount of money has been spent by now, it has become imperative to produce a successful movie. A presumed straight road to success has just become curved for production studios. Hence, it has been learned that there is no shortcut to making a successful motion picture just by investing expensive budget and by casting a reputable star. This problem gave a good opportunity, especially, to computer scientists. It all started with the development of recommendation and predictive software to solve the problem. Recommendation software got much attention when Netflix announced a 1-million-dollar prize for increasing the accuracy of its algorithm by 10%. However, the development of predictive or forecasting models has not received that much attention.

Predicting movie's success has extensively been studied by different experts over all the years which include neural network experts, econometricians as well as marketing professionals. In this paper, we have reviewed both the forecasting models and a variety of analyses conducted by different researchers, but our focus remains at forecasting studies. Since the time of prediction really matters in the case of the movie's success, our whole reviewed work falls into two major categories, i.e. pre-production prediction and post-production prediction. Our review also includes an overview of different domain and analysis of the results of recent research studies.

2.1 Econometric-based studies

Econometricians have been trying to find the contributing variable for revenue prediction which is normally achieved through linear regression analysis of the movie's data and investigation of the correlation between different determinants of revenue (Elberse and Eliashberg 2003). A considerable number of studies have dedicated their research to analyse the performances of sequels of different movies and to the contrary to non-sequel movies (Eliashberg et al. 2006). How do stars affect the success of the movies at the box office has remained a fascinating research area (Eliashberg et al. 2006). The most accurate research reported, so far, by econometricians is known as MOVIEMOD (Eliashberg et al.

2000). It is basically a robust model which was able to forecast with an error rate of 10%, driven with the help of Markov Chains. Since this model has been employed to evaluate the market before the distribution of motion pictures in ancillary markets. MOVIEMOD has proved to be an operative tool for predicting success in different markets.

2.2 Operations-based studies

Littman Barry is known as a pioneer in the field of film industry forecasting. According to him, decision-making can be categorized into three main categories, i.e. film creativity, distribution pattern, and cinema arrangement, marketing (Litman 1983). Zhang et al. (2015) used post-production and post-release-based features, which are based on the word-of-mouth data or from the opening weekend of a film. Marshal predicts the number of viewers of the first week and then used that result to make a prediction of the total viewer in the coming week. His model used prediction of the first week in order to adapt to the nature of the market (Marshall et al. 2013). Backpropagation neural network can also perform better on prediction and classification (using first five-week data) so it provides reliable suggestions for risk assessment before the movie is released (Ghiassi et al. 2015; Sharda and Delen 2006; Delen and Sharda 2010). Internet big data and social media for the box office are increased rapidly. Impact of reviews has an effect on the motion picture success. According to research, the third week after release, the impact of online movie review sentimental orientation on movie income is significant. The negative effect of 1-star review exceeds the positive effect of 5-star reviews, but 2–4 star reviews have no significant impact (Mestyán et al. 2013). Recently, Google released a white paper, according to it on average filmgoers consult 13 sources before they make a decision about what film to see. Moreover, it described the film prediction model depending upon the searches, advertising clicks and a number of screens (Madlberger 2014). The success mainly depends upon its popularity and it can be predicted before its release by measuring and analysing the activity level of editors and viewers of the corresponding entry to film in Wikipedia (Mestyán et al. 2013). Film 40% incomes are generated from the first weekend of theatrical release Einav (2007). A movie revenue is increased linearly from the collection of its first week. Distribution patterns are highly regular (Zhang et al. 2012). The trend of the movie to be successful is highly dependent upon the distribution and promotion strategy. Sometimes an average movie generated a healthy revenuer due to its distribution and promotion strategy (Liu et al. 2016; Chen et al. 2013). To date, there is no forecasting model which effectively predicts the patterns without using a post-release or post-production data. The feature set is very important by considering time constraints (Sharda and Delen 2006; Delen and Sharda 2010; Mestyán et al. 2013). Eliashberg et al. (2007) proposed a model which depends upon the content of the movie, i.e. screenwriting and natural language processing techniques to predict the success of the movie. Without using partial or post-release information, it is very difficult to predict the success of the movie (Altuntas et al. 2015; Liu and Xie 2019).

2.3 Word-of-mouth or social media studies

Social media or word of mouth has been a focal point for the researcher to assess the predictive value of different social media signals. WOM has its own importance when it comes to understanding consumer behaviours for different products. Several studies have tried to apprehend how socially generated data from consumers help with predictive production performance and how they are correlated with performance or sales. Results show that various social media signals have a significant predictive value to predict performance in the box office. A recent study Vosoughi (2015) has concluded that the number of followers of an actor on Twitter has much predictive power for the movie success on box office.

Regardless of the unpredictable nature of the movie domain, quite a few researchers have tried to develop better models to predict the success of the movie in the box office. Researchers who have been trying to build forecasting models are inclined to employ statistical methods at the post-production level. The results were not quite accurate a few years back at pre-release level (Ghiassi et al. 2015; Sharda and Delen 2006; Delen and Sharda 2010). Empirical studies have proved that the preponderance of the revenue comes from the first week of the theatrical release of the movie. This solution gave another direction to researchers to predict movies success in the first week of release with very high exactitude. So, predicting movies success at a pre-production level remains an intriguing problem for the researcher (Ma et al. 2019; Lee and Choeh 2018).

Movie success forecasting models are normally classified into further two categories based on the effectiveness of inputs available to forecast. Furthermore, these studies can also be divided according to the methodologies used. Regression, Bayesian and artificial neural networks models have constantly been employed by the researcher for forecasting the revenue of movie on box office. Forecasting studies which try to predict movies success after its preliminary release are more likely to make more accurate predictions due to several factors such as more parameters, number of receipts, critics, and user reviews (in case of WOM). One of the most notable studies in research of revenue forecasting was Bayesian methods (Neelamegham and Chintagunta 1999). They successfully demonstrated their model by surpassing the accuracy of the model (Sawhney and Eliashberg 1996) 45–71% at the pre-release phase. The artificial neural network generally termed as the neural network has been explored for forecasting research over the years in different domains. Neural network has demonstrated superiority with the counterparts such as regression-based models, discriminate analysis, and support vector machine (Sharda and Delen 2006; Delen and Sharda 2010; Zhang et al. 2009). Post-production forecasting studies normally show better results than pre-production predictions, but techniques to save cost are very critical.

2.4 Forecasting studies

The very first pioneering research in this domain was conducted by Sharda and Delen (2006). This study was the first effort of its own kind, which converted the forecasting problem from simple point estimation into classifications by defining ten different revenue classes. The basic aim of this research was an evaluation of neural network competence to predict a movie's revenue based on the number of explanatory variables. A factor which made this study more contributing to the area was the pre-release prediction. According to the results, reported by this study, using a neural network can predict the revenue with 37% accuracy. The overall accuracy was not promising but compared to other models, neural network significantly performed better than discriminate analysis, logistic regression, and classification trees. As accuracy did not cross a certain threshold which left the window open for future researchers, someone may consider an extension to model, different parameters, and better training and testing. In this study, our focus is to increase the accuracy of Sharda et al. and Delen et al. model with different parameters and features set (Delen and Sharda 2010).

We analyse how efficient artificial neural network has been demonstrated in the above-mentioned study, and changing the algorithm or technique can help to increase the accuracy or not. The following study employed Backpropagation neural network (Zhang et al. 2009), unlike the Sharda et al. which employed multi-layer perceptron to predict movie financial performance of box office (Sharda and Delen 2006; Delen and Sharda 2010). Movies were divided into six different classes according to the revenue they earned. The number of layers of training the model went through several experiments before it was accepted which yielded a better predictive model. Normally, 10-fold cross-validation is employed to split the data for training and testing as it has become a standard methodology, but changing the split ratio helped to achieve optimal performance. BP neural network for predicting financial performance has shown a statistically significant improvement of 68.1%, which was previously only 37%. Hence, BP neural network compared to the MLP neural network is credible to build predictive models for the movie domain.

Table 5 Sharda and Delen variables and ANN inputs

Feature	Possible value
MPAA rating	G, PG, PG-13, R, NR
Competition	High, medium, low
Star value	High, medium, low
Genre	Sci-Fi, epic drama, modern drama, thriller, horror, comedy, cartoon, action, documentary
Special effects	High, medium, low
Sequel	Yes, no
Screen count	A positive integer between 1 and 3876

2.5 Artificial neural network based forecasting model

Sharda and Delen used the pre-production variant which includes MPAA rating, competition, star value, genre, special effects, sequel, and the number of screens. MPAA ratings categorized as G, PG, PG-13, R, and not-rated as mentioned in Table 5. Competition, star value, and special effects are also further categorized into low, medium, and high values. The sequel variable is classified into binary number 0 or 1. The number of screens is given a numeric number depending upon the number of screens at the times of its release.

2.5.1 Feature set

Sharda and Delen include seven parameters in their feature set. MPAA rating is generally assigned a week before movie release and describes the audience size (Terry et al. 2011). Basuroy used the regression analysis and find that MPAA rating has a very strong relation to revenues (Basuroy et al. 2003). It is a general trend that action and violence genre movie performance is better in term of revenue generation (Dahl and DellaVigna 2009). Genre and date of release have a strong correlation with revenue (Prieto-Rodriguez et al. 2015). Season of release and competition plays a critical role in movie success (Jedidi et al. 1998; Terry et al. 2010; Galvão and Henriques 2018). The date and time of movie release play an imperative role to generate better revenue than the average movie. Movies released in parallel or alongside play a negative impact on the success of the movie because of market share (Sharda and Delen 2006; Delen and Sharda 2010). Delen and Sharda considered the month of release for categorizing parameter competition to be high, medium and low. They measure star value by examining the star presence or absence in the movie. They also categorized star value further into high, medium and insignificant classes. They classify each movie into ten numeric numbers by considering its genre, i.e. science fiction, historical epic drama, modern drama, politically related, thriller, horror, comedy, cartoon, action, and documentary. Technical effects parameter is also used in their study. They rated each movie high, medium, and low by measuring its technical content. The sequel is another parameter which is included in their data set and given a binary number 1 and 0. Due to brand awareness by parent films, sequels perform better than the average movie (Hennig-Thurau et al. 2009). Rotten Tomatoes and Yahoo movie found that sequel can attract larger audiences, and get lower ratings than its first version (Moon et al. 2010a). Cinema-goers can predict the story or assumes the type of film as a result sequel tends to outperform non-sequels (Moon et al. 2010b). The number of screens and revenue generated by the film has a strong positive correlation (Neelamegham and Chintagunta 1999; Basuroy et al. 2003; Moon et al. 2010b).

2.5.2 Analysis of the feature set

The MPAA rating is assigned by the Motion Picture Association of America. The rating is assigned only if they watch the complete movie so by selecting this feature we must wait for the rating of the movie to be assigned after that model will predict. As described earlier that prediction is of no value as studio and investor already put its money in the movie. Same in the case of competition, we must wait for the release date of the movie. So, media houses or reporter only reports that the movie is released in specific when the movie is almost or 80% completed. So, by getting this value and making the prediction will not benefit investor or verdict maker. Screen count feature value is only getable as we know the released date and by getting released its mean money of the investor is already being spent so prediction at that stage have some value but cannot add much in decision-making. In their ANN-based approach, they assigned each movie a class from a blockbuster to flop by defining ranges of each class as mentioned in Table 6. In this way, the forecasting problem is converted into a classification problem.

Table 6 Sharda and Delen: classification thresholds

Class	Revenue range (in $ millions)
(Blockbuster) A	200 +
B	150–200
C	100–150
D	65–100
E	40–65
F	20–40
G	10–20
H	1–10
(Flop) I	< 1

2.5.3 Performance measure

In the classification problem, average percentage hit rate (APHR) is the ratio of correctly classified labelled and total number classes. Sharda and Delen introduced two different hit rates, i.e. the exact (bingo) hit rate (Ghiassi et al. 2015; Sharda and Delen 2006; Delen and Sharda 2010) (Table 7).

$$\text{APHR} = \frac{\text{Number of Sample Correctly Classify}}{\text{Total Number of sample}} \tag{1}$$

$$\text{APHR}_{\text{bingo}} = \frac{1}{n} + \sum_{i=1}^{g} P_i \tag{2}$$

$$\text{APHR}_{1-\text{away}} = (P_1 + P_2) + \sum_{i=1}^{g} P_{i-1} + P_i + P_{i+2} + (P_{g-1} + P_g) \tag{3}$$

where

G is the total number of classes.
N is the total number of samples.
P is the total number of samples classified as class.

2.6 Study on dynamic artificial neural network

Ghaissi presents a forecasting model using dynamic neural networks during the pre-production period (Ghiassi et al. 2015). Their model improves the forecasting of Sharda and Delen by 32.8%. They also offer an alternative modelling strategy by adding production budgets, pre-release advertising expenditures, runtime, and seasonality to the predictive variables. The model produces the forecasting accuracy of 94.1%.

Table 7 Delen and Sharda (2010): prediction models and results

Performance measure	SVM	ANN	CART	Random forest	Boosted tree	FUSION average
Count (Bingo)	192	182	140	189	187	194
Count (1-Away)	104	120	126	121	104	120
Accuracy (% Bingo)	55.49%	52.60%	40.46%	54.62%	54.05%	56.07%
Accuracy (% 1-Away)	85.55%	87.28%	76.88%	89.60%	84.10%	90.75%

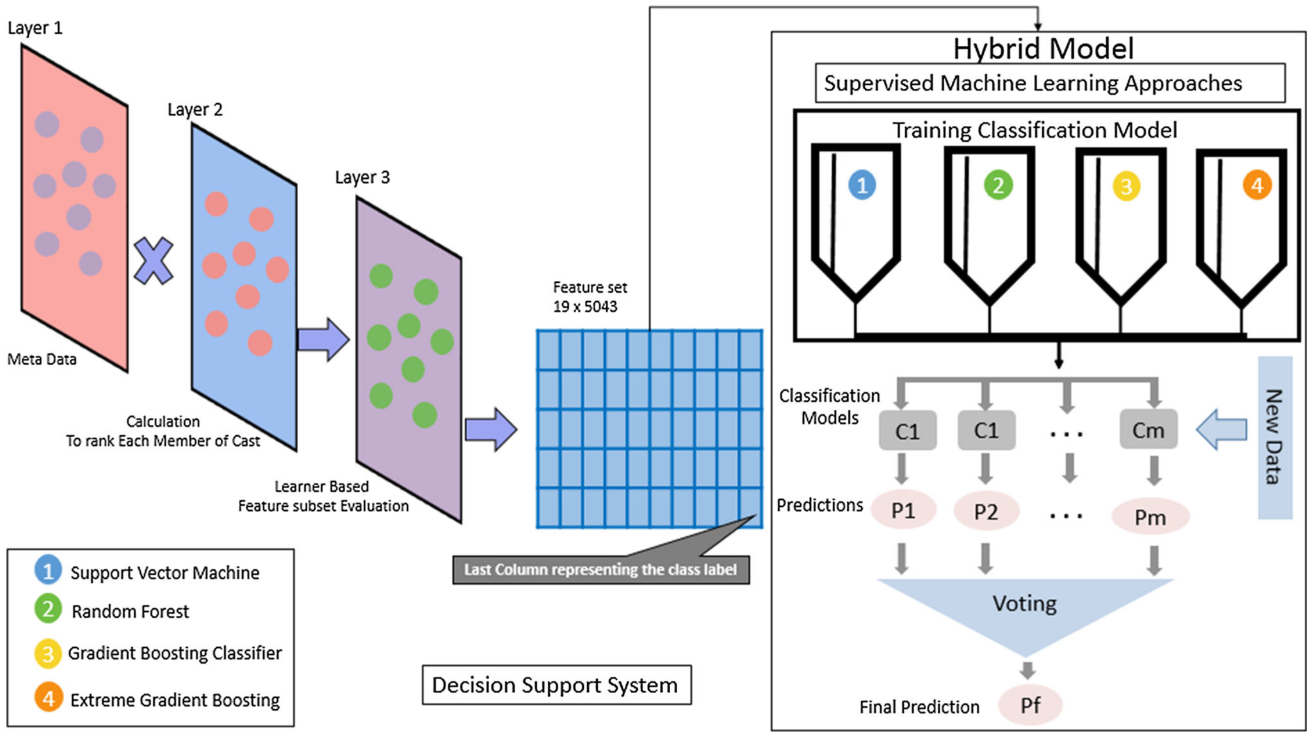


Fig. 1 System overview

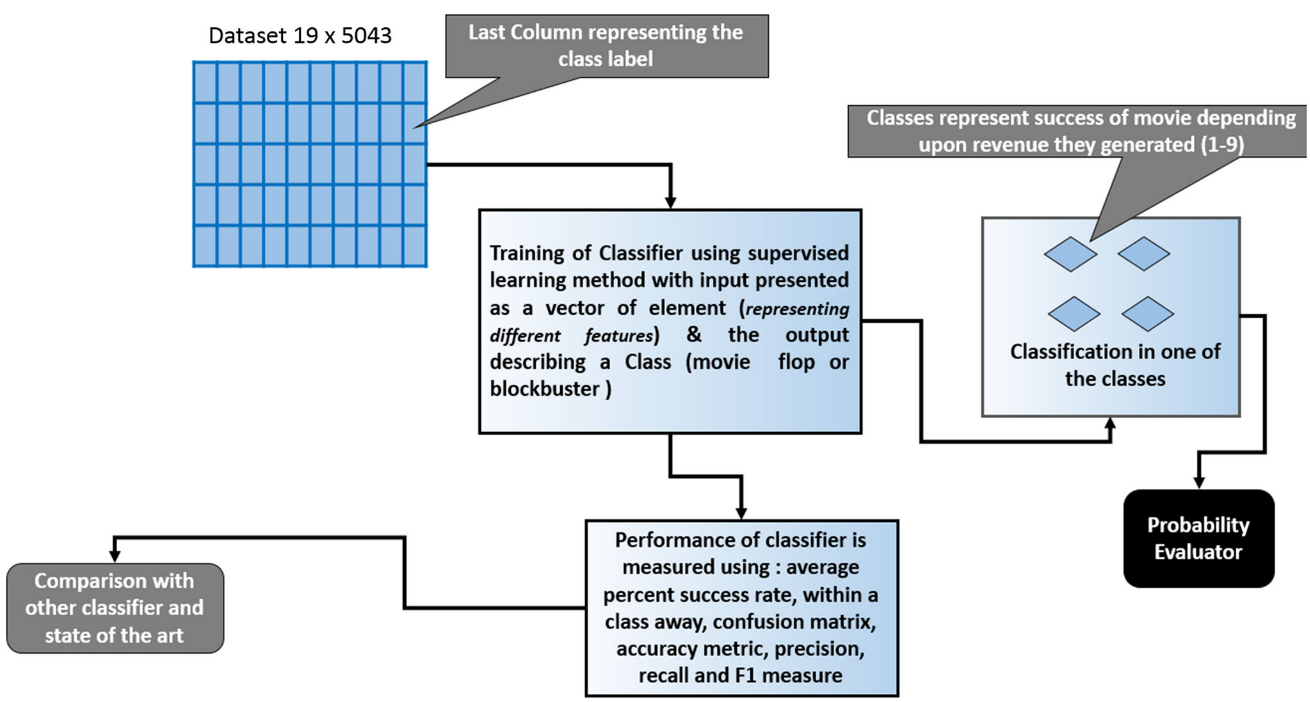


Fig. 2 Model overview

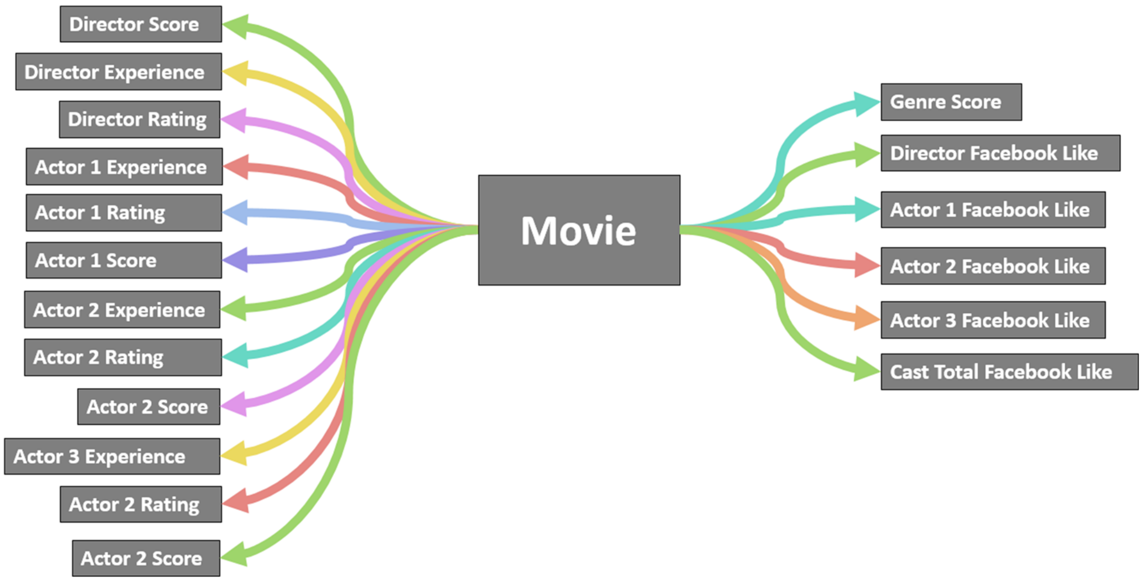


Fig. 3 Time independent feature set

2.6.1 Analysis on feature set

Their feature set includes production budgets, pre-release advertising expenditures, runtime, and seasonality. As described earlier that prediction has no value when studio and investor have already spend their money in the movie. Production budget value is only getable when the story, director, and cast are finalized. The runtime of the movie is known when the movie is completed and reached its editable process. When a movie is in the production phase, there is no concrete reasoning when it will end and ready for an audience. So, these features set are used for post-production prediction. Pre-release prediction is only possible when we have historical data of cast (director, lead actor, or actress) past performances and their reputation in cinema goer's audience.

Methodology

There is no general approach to assert the integrity of the films. Many individuals depend on critics to gauge the nature of the film, while others utilize forecasting insight, but it takes time to obtain a reasonable amount of data after or before a movie is released. Moreover, human instinct some of the time is untrustworthy. Given that a huge number of motion pictures were delivered every year, is there a superior path for us to tell the enormity of motion pictures without depending on pundits or our own impulses?

The focal point of this research is to analyse how different features can be used to predict the success of the movie. Initially, a data set is collected that later trains the grading machine learning model. The data set contains movies information that releases the last 100 year (1915–2015). Using this corpus, a data set comprising of nineteen columns and 5043 rows is created as mentioned in Figs. 1 and 2. The last column of the data set is the class label, i.e. the output class. The data is classified in one of the nine classes, each represents a particular classification range as used by Sharda and Delen (2006), Delen and Sharda (2010). The data set is pre-processed to remove the irregularities and to provide uniformly important input variables to the model. The hybrid model is trained using a supervised learning method with input presented as a vector of elements, yielding one of the nine blockbuster or flop class explained in Table 6. The classifier's results are validated using the test data of 2016 movies. The system components and its structure are shown in Figs. 1 and 2. The rest of this section discusses the data set built for the proposed system.

Data set acquisition

The data is acquired from NYC Data Science Academy (Ma et al. 2019), which was uploaded on Kaggle (an open source data sharing website).

The python script was used to get the last 100 years of movie data. The python script tool is used to extract data from the IMDB website as mentioned in Fig. 1. They used the Python library named" scrappy" to obtain a list of 5043 movie titles from" the-numbers" website, which was saved into a JSON file format. Then, tool searches those titles from the IMDB website to get the real IMDB movie links. After that, it sends HTTP request to each movie page using the links, and scrappy get all data. Then, it parses the aggregated data, cleans it, and reformats it to a CSV file.

A data set is formed based on the knowledge available about each movie sample. The metadata consists of: director name, $actor_1$ name, $actor_2n$ name, $actor_3$ name, movie genre , director Facebook likes, $actor_1$ Facebook likes, $actor_2$ Facebook Likes, $actor_3$ Facebook likes, cast total Facebook likes and the relevant task to be carried out. This paper focuses on the tasks to calculate the score, experience, and rating of the director and cast. Nine ranges of revenue are considered to construct the data set which becomes nine classes to be considered by the classifier. These classes are labelled as: (1–9) each having director experience, director rating, $actor_1$ experience, $actor_1$ rating, $actor_1$ score, $actor_2$ experience, $actor_2$ rating, $actor_2$ score, $actor_3$ experience, $actor_3$ rating, $actor_3$ score, genre score, director Facebook likes, $actor_1$ Facebook likes, $actor_2$ Facebook likes, $actor_3$ Facebook likes, cast total Facebook likes samples, respectively. The numbers of feature occurring in a particular class remain the same since they depend on their usefulness in contemporary work. The data set is mathematically shown in Eqs. (4)–(11) which shows the mapping between metadata and the tasks. D represents a set of metadata where each d representing different Feature, T is a set of tasks and C is the set of classification. Later, a relation is found between D and T using a suitable mapping in set C. A simple instance of the data set is dimension $= 1$, number of attribute $= 1$. The instances are 5000 or less and given the distribution task is to map to the target classification of the movie (blockbuster or flop).

$$D = \{d : d \in \text{ metadata}\} \tag{4}$$

$$T = \{t : t \in \text{ task on data (Calculation)}\} \tag{5}$$

$$C = \{c : c \in \text{ Classes (Depending upon revenue range)}\}. \tag{6}$$

then,

$$Dx\ T \rightarrow c = \langle d, t, c\rangle \tag{7}$$

$$D = \langle d_1, d_2, d_3 ....d_{18}\rangle \tag{8}$$

$$d = \langle a_1, a_2, a_3 ....a_{18}\rangle \tag{9}$$

The task set T is represented as

$$T = \langle t_1, t_2, t_3 ....t_{18}\rangle \tag{10}$$

Similarly, For classification we have the set C,

$$C = \langle c_1, c_2, c_3 ....c_{18}\rangle \tag{11}$$

2.7 Features set

About 5043 essential film data is considered and scratched from the IMDB website. In general, the data set contains more than 100 years of motion picture information, which was released in 66 countries. There are 2399 unique director names, and 30,000 performing artists/on-screen characters. Figure 3 demonstrates all the 28 factors that were scratched. Generally, half of the factors are specifically identified with motion pictures themselves, for example, title, year, length, and so forth. Another half is identified with the general artist population who required in the generation of the films, e.g. directors names, directors Facebook popularity, motion picture rating from pundits. The response variable in this research is profit. This research follows the footstep of pioneering research work in the domain defining the class of each profit. Assignment of profit ranges with classes is mentioned in Table 6. We proposed 18 unique features which contain the last hundred-year data. Such diverse data set is not used in the movie forecasting literature. Moreover, this feature set is not dependable on any media information. This featured model can make a prediction on the day cast is finalized and an agreement is signed. The detail about the feature and how this feature variable is calculated is mentioned in the table. The proposed model is trained on the last 100 year (1915–2015) data and makes the prediction of 2016 movies (Tables 8 and 9).

2.7.1 The internet movie database (IMDB)

The Internet Movie Database (abbreviated IMDB) is an online database of information related to films. IMDB assigns ratings of movies as soon as the movie trailer is released. They assign rating according to this formula mentioned Eq. (12). The rating is used to give actor rating according to film they worked on.

$$W = \frac{\text{Total score attained}}{\text{Number of users voting}} \tag{12}$$

Class of movie is also classified as mentioned Table 10. By reviewing the last 100 year, this research classifies each genre of the movie into the main seventeen categories which were mentioned below.

2.7.2 Facebook

Nowadays, social marketing is one of the top advertisement components. According to the source (Galvão and Henriques 2018), 84% trust recommendation from family, colleagues, and friends about any product. 68% of people get influenced by the other consumer, making social network trusted source. Word of mouth is the key influencer in their purchasing decision. 74 % of people follow the trends in the social network (Lee and Choeh 2018). Usually, in social network place, peoples comments on different product and about 88% trust online reviews as they trust recommendations from personal contacts (Liu and Xie 2019). 43% Facebook users

Table 8 Feature set

S. no	Feature	Formula
1	Director score	Sum of revenue collected by its film/experience
2	Director experience	Number of film directed
3	Director rating	Sum of IMDB rating given to directed movie/experience
4	Actor 1 experience	Number of film acted
5	Actor 1 rating	Sum of IMDB rating given to directed movie/experience
6	Actor 1 score	Sum of revenue collected by its film/experience
7	Actor 2 experience	Number of film acted
8	Actor 2 rating	Sum of IMDB rating given to acted movie/experience
9	Actor 2 score	Sum of revenue collected by its acted movie/experience
10	Actor 3 experience	Number of film acted
11	Actor 3 rating	Sum of IMDB rating given to acted movie/experience
12	Actor 3 score	Sum of revenue collected by its acted movie/experience
13	Genre score	Sum of revenue collected by specific genre/experience
14	Director Facebook likes	Director official Facebook like
15	Actor 1 Facebook likes	Actor 1 official page Facebook like
16	Actor 2 Facebook likes	Actor 2 official page Facebook like
17	Actor 3 Facebook likes	Actor 3 official page Facebook like
18	Cast total Facebook likes	Cast total official page Facebook like

Table 9 Feature set sample

Director score	Director experience	Director rating	Actor 1 experience	Actor 1 rating
31,439,603	7	6.228571	29	6.913793
62,476,106	5	5.64	1	7.5

Actor 1 score	Actor 2 experience	Actor 2 rating	Actor 2 score	Actor 3 experience
61,087,552	2	7.1	38,770,097	1
52,418,902	2	7.1	38,770,097	1

Actor 3 rating	Actor 3 score	Genre score	Director FB likes	Actor 1 FB likes
6.7	25,121,291	76,654,271	101	13,000
7.5	52,418,902	30,173,094	845	15,000

Actor 2 FB likes	Actor 3 FB likes	Cast total FB likes
1000	585	15,708
1000	424	16,881

share and report buying a product. There are 20,000 people on Facebook every second. That number alone makes it a valuable website for marketers and advertisers. Facebook is ranked top among the other social networks, with 1.44 billion monthly active users, 1.25 billion mobile users, 936 million daily active users, 798 million mobile daily active users (Sprout Social 2016). 27% adults age between 18 and 29 have a social network of more than 500 people, while 15% of 30–49 years do (Pew Research 2016). Advertisers and marketers are mostly looking for a larger target audience within a short period of time, so Facebook provides a large window of time to work on Pew Research (2016). Figure 4 explains Facebook has a user who spends more as compared to other networks. By considering the target audience, Facebook is playing a significant role in term of promotion prospects. User likes page of the favourite movie cast and wants to know more about them. So casts Facebook page like is also added to the database. These features are the director, first, three lead casts, and movie Facebook page likes are collected.

2.7.3 Country-wise movie production

After analysis, we can say that the USA delivered the biggest measure of motion pictures over the previous 100 years (1905–2015) (Ghiassi et al. 2015). Numerous nations cre-

Table 10 Genre classification

S. no	Genre
1	Action
2	Adventure
3	Animation
4	Biography
5	Comedy
6	Crime
7	Documentary
8	Drama
9	Family
10	Fantasy
11	Horror
12	Musical
13	Mystery
14	Romance
15	Sci-Fi
16	Thriller
17	Western

ated awesome films and profitable, yet at the same time, there were many terrible motion pictures in term of profit. Motion pictures having rating bigger than 8.0 are recorded in the IMDB best 250, and they are genuinely extraordinary motion pictures from numerous point of views. Films with a rating from 7.0 to 8.0 are likely still great motion pictures. Films rated less than 5 should be avoided watching. The USA and UK are the two nations that created the most number of films in the previous century, including a lot of terrible motion pictures in term of profit. The middle IMDB scores for both the USA and UK are, in any case, not the most astounding among all nations. Some movie creating nations, for example, Libya, Iran, Brazil, and Afghanistan, delivered a little measure of films with high middle IMDB scores. In the most recent century, it appears that the number of motion pictures created yearly generally expanded since 2010. This is justifiable for the advancement of the shooting industry runs as one with the improvement of science and innovation. Be that as it may, we ought to know that alongside the blast of motion picture industry since 2000, there are numerous films with low IMDB scores as mentioned in Fig. 5.

2.7.4 Facebook page likes

As described earlier, Facebook promotion plays a huge impact on the success of the movie. From the dissipate plot underneath, we can find that by and large, the motion pictures that have high Facebook likes have a tendency to be the ones that have IMDB scores around 8.0

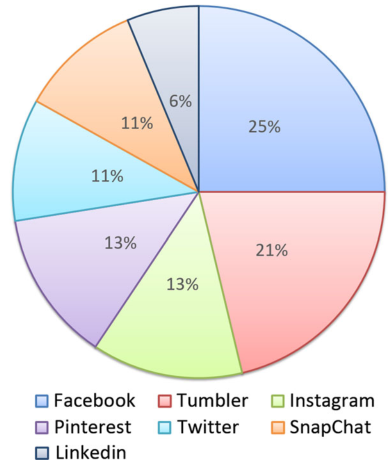


Fig. 4 Average time spend on social network

2.7.5 Director and actor features

It is conceivable to trust that the enormity of a motion picture is exceptionally influenced by its direction. Figure 8 expresses the motion picture IMDB score with director Facebook prevalence. Low Facebook likes have a tendency to have low rating, whereas high page likes have high appraising. Awesome performers/on-screen characters make a motion picture extraordinary. They are the souls of motion pictures. The reason to explore this feature is that we want to analyse that how does their Facebook fame resemble? For a given movie, this research scratched all the accessible cast individuals in the IMDB film page. After recovering the quantity of Facebook preferences for all cast individuals, position the numbers in dropping the request and picked the main 3 performers/on-screen characters. This depends on a straightforward presumption: driving on-screen character/performer has a tendency to have more Facebook fame than supporting on-screen character/on-screen character; and regardless of how awesome a motion picture is, there will be close to 2 driving on-screen characters/on-screen characters. In the same way, they generated a higher score by formula mentioned in Table 8. Likewise, highest rank director and actor have a high percentage of award nomination and winning as mentioned Tables 11, 12, Figs. 7 and 10. For documentation space, the Facebook prevalence for the main 3 performing artist/performer as “actor 1 Facebook likes”, “actor 2 Facebook likes”, and “actor 3 Facebook likes” is used. Note likewise that the variable “cast total Facebook likes” is ascertained by summing up the Facebook prevalence of all the accessible cast individuals. The presumption in reality coordinates with the plotted chart beneath. The top first performer/on-screen character has the most number of Facebook prevalence, while the second and the third on-screen character/on-screen character have much lower ubiquity. In any case, it can likewise be demonstrated that the high Face-

Fig. 5 IMDB rating comparison with profit and number of movie

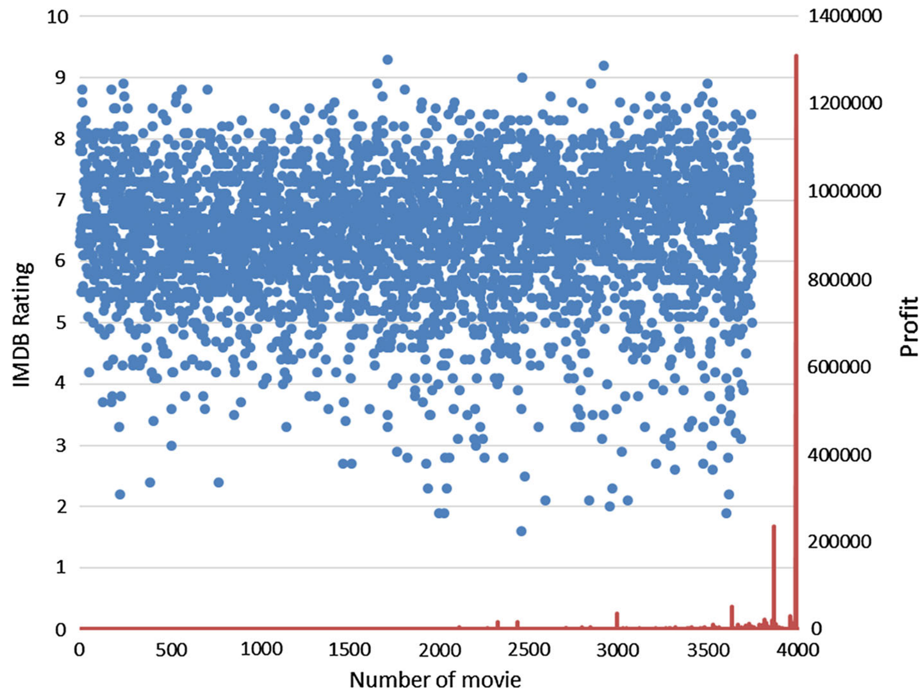


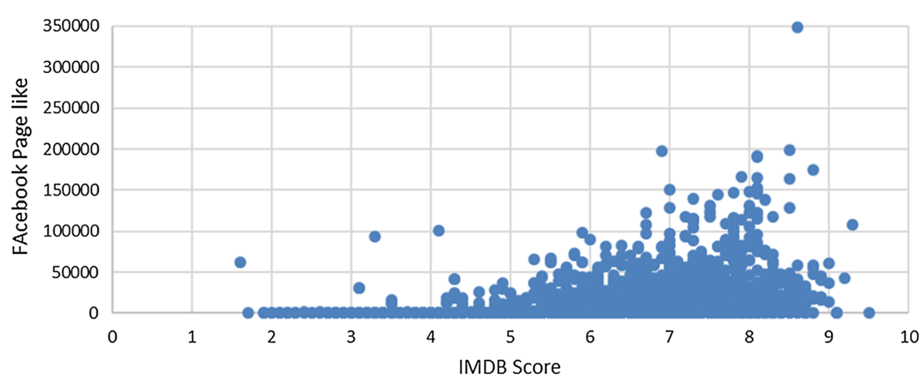
Table 11 Director ranking with score formula

Rank	Name	Number of award nomination	Number of awards won	Oscar
1	Joss Whedon	40	20	Nominated
2	Lee Unkrich	24	15	Won
3	Chris Buck	10	10	Won and nominated
4	Tim Miller	7	1	Nominated
5	George Lucas	32	54	Won

Table 12 Actor ranking comparison with score formula

Rank	Name	Number of award nomination	Number of awards won	Oscar
1	Bruce Willis	37	22	Nomination
2	Tom Hardy	72	16	Nomination
3	Christian Bale	81	64	Nominated and won
4	Tom Cruise	77	46	Nominated and won
5	Will Smith	112	56	Nominated

Fig. 6 Facebook page like comparison with IMDB score



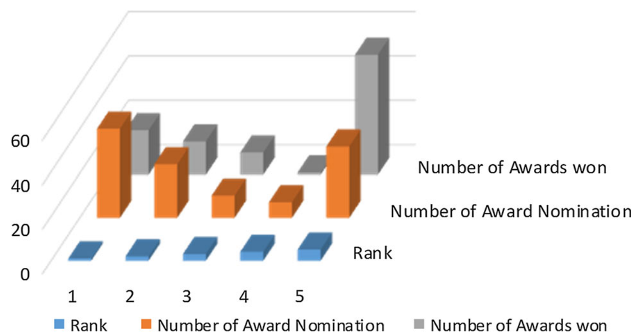


Fig. 7 Director ranking comparison

book fame of the main performing artist/on-screen character does not imply that a motion picture is of high appraising.

Machine learning model

In this study, supervised machine learning model implemented is a meta-classifier by combining conceptually different machine learning classifiers for classification. A voting system is adopted to predict the class labels by taking the average of the output class as mentioned in Figs. 1 and 2. Basically, our data set has consisted of examples and in a pair of X, Y, where X is called a feature vector which gives the factorial description of the objects which we want to generate predictions as in our case Director Rank, is one feature and seventeen others. On the other hand, Y is the response which in our case is 1–9 different classes and mostly called a classification problem. The goal of supervised machine learning is basically to find the function that is mapped (X, Y) in such manner that the error is minimum for unseen data, which is measured by loss function or with accuracy rate (Fig. 6).

$$(X_i\ ,\ Y_i\ ),\ 0\ \leq\ i\ <\ \text{number of samples} \tag{13}$$

$$X \in R^{n\text{features}} \tag{14}$$

$$Y \in 1, 2, 3, 4, 5, 6, 7, 8, 9\ \text{ classification} \tag{15}$$

$$Y^{\circ}\ =\ F(x) \tag{16}$$

Such that $L(Y, Y^{\circ})$ or new(unseen) x is minimal

Gradient boosting

Grading boosting technique is used by major search engine companies, i.e. Google, Bing, Yandex, and Yahoo. They use it for web page ranking, but it is actually not limited to the domains and can be used for a variety of problems (Viola et al. 2001). Gradient boosting classifiers are models made up of different weaker models that are trained individually, and each models prediction is combine. The combination of a weak model is an effective strategy. This is an effective strategy and accordingly is extremely famous (Figs. 7, 8, and 9).

Gradient boosting is a standout among the most powerful techniques for building classification models. The idea is to combine weak leaner in such a way that overall model accuracy is optimal. The output of different weak learner is combined in such a way that its loss function should be optimized. The main objective is that each model loss should be minimized by adding weak learner using procedures like gradient descent. When one weak leaner is added in the model, the other left unchanged. Gradient boosting involves three elements (Fig. 10):

1. A loss function to be optimized.
2. A weak learner to make predictions.
3. An additive model to add weak learners to minimize the loss function.

2.8 Extreme gradient boosting

Both extreme gradient boosting and gradient boosting follow the principle of gradient boosting as mentioned above. There is, however, the difference in modelling details. Specifically, extreme gradient boosting used a more regularized model formalization to control over-fitting, which gives it better performance.

Before running extreme boosting classifier (Xgboost), we must set three types of parameters: general parameters, booster parameters, and task parameters. The detail about each parameter can be accessed from their page which has very rich information and that help us a lot (Xgboost 2016; Zafar et al. 2017; Ahmed et al. 2019, 2018).

1. General parameters relate to the booster we are using for boosting, commonly tree, or linear model
2. Boosting parameters depend on which booster you have chosen.
3. Learning task parameters that decide on the learning scenario, for example, regression tasks may use different parameters with ranking tasks.

$$\sum_{k=1}^{k} f_k \tag{17}$$

$$Y_i\ =\ \sum_{k=1}^{k} f_k(x_i) \tag{18}$$

$$Y_i^{(t)}\ =\ \sum_{k=1}^{t} f_k(x_i) \tag{19}$$

$$\text{Obj}(Y, Y^{\circ}) \tag{20}$$

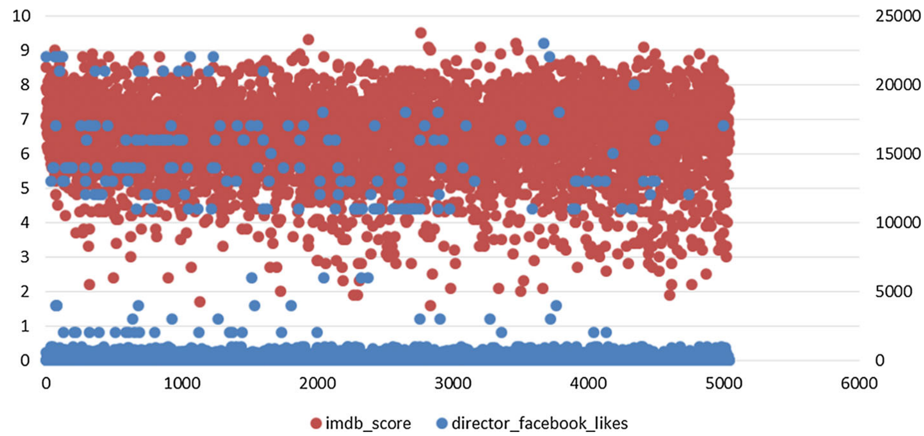


Fig. 8 Director Facebook page like comparison with IMDB score

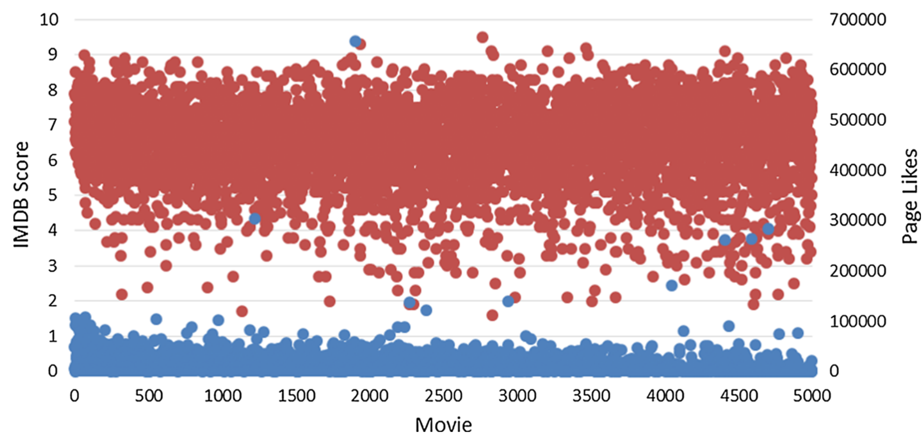


Fig. 9 Cast total Facebook page likes

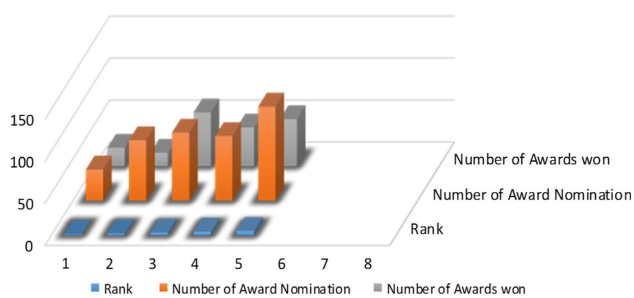


Fig. 10 Actor rankings

2.8.1 Max depth

Maximum depth of a tree, increasing this value will make the model more complex/likely to be over-fitting. The default value is 6, but we set it to 10, according to our feature set as mentioned in Table 5. The maximum depth limits the number of nodes in the tree. The maximum depth parameter is required to be tune according to provided data and the best value.

2.8.2 Learning rate

The learning rate is mostly set to average, but if we set it too small then the model needs more time to converge and if set to large value then it will not perform well but trained in less time. The learning is kept small 0.06 so that model should be a good learner as mentioned in Table 13. Learning rate shrinks the contribution of each tree by $\text{learning}_{\text{rate}}$. There is a trade-off between $\text{learning}_{\text{rate}}$ and $n_{\text{estimators}}$.

Table 13 Extreme gradient boosting parameters

$\text{Learning}_{\text{rate}}$	$\text{Max}_{\text{depth}}$	$\text{Min}_{\text{childweight}}$	$n_{\text{estimators}}$	Subsample
0.06	10	1	500	1

2.8.3 Minimum child weight

The minimum sum of instance weight needed in a child. If the tree partition step results in a leaf node with the sum of instance weight less than $\text{min}_{\text{childweight}}$, then the building process will give up further partitioning. By default, it is 1.

2.8.4 Subsample and estimators

It is the subsampling ratio of the training instance. Setting it to 0.5 means that boost randomly collected half of the data instances to grow trees and this will prevent over-fitting. The range is (0, 1) and we set it to 1. The estimators are the number of boosting stages to perform. Gradient boosting is fairly robust to over-fitting so a large number usually result in better performance.

2.9 Random forest

Random forest is an adaptable machine learning strategy equipped for performing both regression and classification tasks. It is a type of ensemble learning method, where a weak tree model combines in a manner to form a powerful tree model. During the classification of a new object, each tree model gives a classification and each tree classification is taken into account. After that, finalized decision is made by taking the average of the different tree (Khalid et al. 2019; Svetnik et al. 2003; Zafar et al. 2017; Ahmed et al. 2019, 2018).

2.10 Support vector machine

Support vector machines are perhaps a standout among the most well-known and discussed machine learning algorithms. It remains in mainstream around the time they were created in the 1990s and keep on being the go-to technique for a high-performing algorithm with the little tuning. It is a discriminative classifier, given labelled training data (supervised learning), the algorithm outputs an optimal hyperplane which categorizes new example. On the basis of this training, the algorithm is able to predict unknown input. In this system, we used one versus rest classifier. The general approach is to split the input variable space with a line. The distance between the line and the data point is called a margin. The line said to be the optimal line or best when it has the largest margin. This is called the maximal margin hyperplane. The margin is calculated as by the distance from the line to only the closest points. These points are called the support vectors. These vectors define the hyperplane. It learned from training data using an optimization procedure by maximizing the margin. The SVM algorithm is implemented in practice using a kernel. There are different kernels linear, polynomial, but we implemented radial basis kernel function can be described by Eq. (21).

$$K(x, x_i) = \exp(-gamma * sum(x, x_i)) \tag{21}$$

where gamma ranges between 0 and 1. The radial kernel creates complex regions within the feature space.

2.11 Voting

The voting mechanism is adopted which takes account of each model class prediction, and then, average is taken to predict the final class of model. Equation (22) explains it briefly, where w_j is the weight that can be assigned to the jth classifier. The section information is taken from this source Scikit (2016).

$$Y = argmax_i \sum_{j=1}^{m} w_j \, p_{ij} \tag{22}$$

Table 14 Testing accuracy

Class	Testing sample	F1 measure
A	32	0.783784
B	18	0.611111
C	56	0.816667
D	63	0.794118
E	120	0.826446
F	129	0.874074
G	91	0.789474
H	106	0.919643
I	112	0.931034
Total	727	0.82

3 Box-office forecasting

We presented a hybrid model, utilizing eighteen proposed features data set of the last 100 year. We used the average per cent success rate, within a class, confusion matrix, accuracy metric, precision, recall, and F1 measure as a performance measure. Feature extraction mechanism introduced in this research has a prime focus on the value which purely depends upon on historical data. By considering time as an important constraint for prediction, the proposed decision support system is able to forecast at the very early stage of the production. Performers measure proves that the new feature set has improved the accuracy than of Sharda and Delen (2006) data set. System higher accuracy proves that by using an appropriate model and feature extraction technique, optimal results can be achieved only by using historical data, i.e. by measuring the cast previous performance and revenue generated by the movie. The features used in this paper are available on the day cast signs the agreement without waiting for features used by Ghiassi et al. (2015), Delen and Sharda (2010), Mestyán et al. (2013), Moon et al. (2010a). In other words, this feature model can predict at the earliest part of the film release, which was never used by any other research (Table 14).

Our classification methodology remained the same as used by Sharda and Delen (2006), Delen and Sharda (2010). We performed experiments to calculate the mentioned performance measure. By using performance measure used by (Sharda and Delen data set), i.e. the average per cent suc-

Table 15 Accuracy metrics

Predicted											
Actual	A	B	C	D	E	F	G	H	I	APHR Bingo	APHR 1-Away
A	29	3	2	2	0	1	0	0	29	0.783784	0.864865
B	3	11	4	0	0	0	0	0	3	0.611111	1
C	0	2	49	5	4	0	0	0	0	0.816667	0.933333
D	0	1	6	54	3	4	0	0	0	0.794118	0.926471
E	0	1	0	8	100	12	0	0	0	0.826446	0.991736
F	0	0	0	5	7	118	4	1	0	0.874074	0.955556
G	0	0	1	0	3	8	75	8	0	0.789474	0.957895
H	0	0	0	1	0	5	3	103	0	0.919643	0.946429
I	0	0	0	0	0	0	4	4	0	0.931034	0.965517
Avg										0.82	0.95

Table 16 Improvement with new features

Classifier	Features	APHR Bingo	APHR 1-Away	Improved result	
Hybrid classifier	New features	0.82	0.95	APHR Bingo	APHR 1-Away
Fusion model (Delen and Sharda 2010)	Old features	0.56	0.9075	46.43%	5.56%
Dynamic ANN (Ghiassi et al. 2015)	Old features	0.71	0.74	15.49%	28.38%

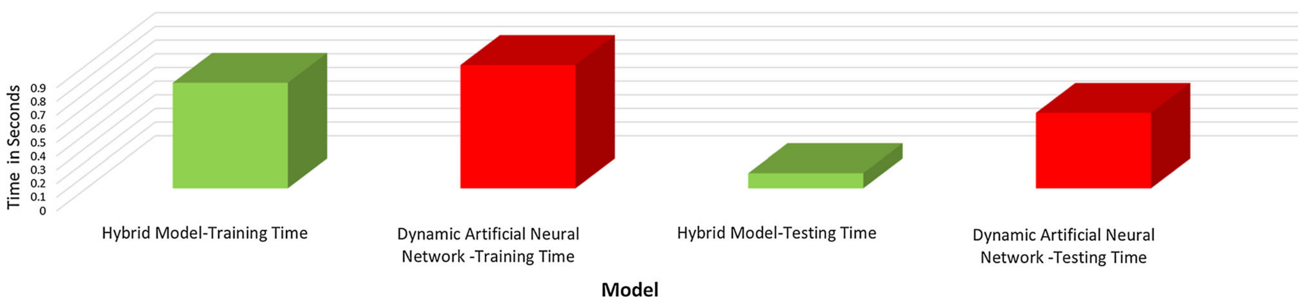


Fig. 11 Computational time * dynamic artificial neural network (Ghiassi et al. 2015)

cess rate and within a class away. The proposed model has a high accuracy rate than the feature used by the Sharda and Delen data set, with a 5.56% improvement in average per cent success and 46.46% within a class away improvement mentioned in Tables 15 and 16. The computational cost of model (Ghiassi et al. 2015) is also very high as compared to the hybrid model as mentioned in Fig. 11. Hybrid model has low training and testing time which makes this model more efficient.

To the best of our knowledge, these prediction results with that early timing are better than any reported in the published literature for this problem domain. Table 16 can show the improvement with other published work. The consistently astounding performance of this model validates the performance of the new feature sets for film industry forecasting.

4 Conclusion

A large amount of investment is spent on every box-office movie. The 90% of the investment even fails to recover the production budget. Previously, models have prime focused on the post-release and post-production forecasting. Many depend on movie critics and news reporter reviews to gauge the nature of the film, while some utilize forecasting insight to check the trend. The prediction has no value as studio and investors have already spent their money on the movie production. The collection of variables like production budgets, pre-release advertising expenditures, runtime, seasonality, and information related to movies, requires a reasonable amount of time. Moreover, forecasts at the later stage are of no value to the decision maker, studio, or investor whose cash has as of now been spent. The pre-production forecast requires high accuracy, as well as the right time to secure

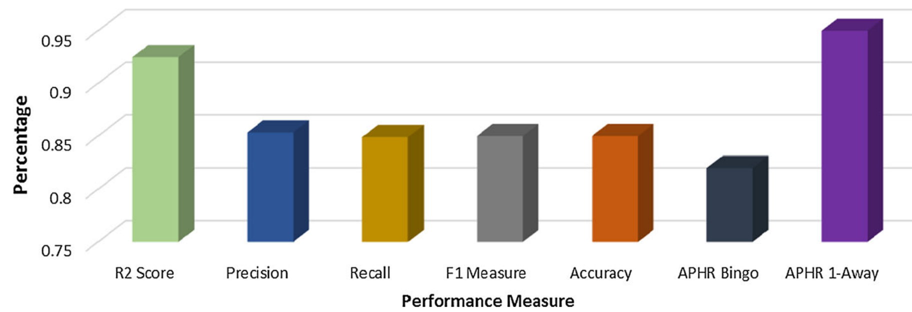


Fig. 12 Results

investment. A forecast made immediately as the cast is finalized, will help the investor to save the investment.

The focal point of this research is to propose a system which can make a prediction on the day when a quotient (director or cast) signs an agreement. To the best of our knowledge, this proposed forecasting time is the earliest prediction that is ever reported in the movie forecasting literature. The data set is gathered from different freely available websites. The movie data set has a collection of 100 years (1915–2016). The collected data set is more comprehensive in nature than those already referred to in the literature. This paper evaluates eighteen components to arrange every film as the blockbuster or a flop. The proposed feature set does not depend upon any media reporting. It uses purely historical data to predict movie success. This research model makes the prediction in the pre-production period when requisite data for alternative models are generally unavailable.

The prime objective of this research is to improve the accuracy of pre-production revenue forecast model proposed by Ghiassi et al. (2015), Sharda and Delen (2006), Delen and Sharda (2010) by using new features. Our goal is to develop a forecasting model which makes the prediction at an early stage of movie production, which has practical value to investors or verdict maker in the film industry. The profiling of 2399 directors and 30,000 on-screen characters is calculated by the proposed formula. The formula is based on the parameter like experience, journalist critics, media reporting, user ratings, and revenue generated by an associated movie. Learner-based feature selection is performed to extract most influence features from the formulated data. The movie classification or output is based on Sharda and Delen proposed model, which classify each movie based on the revenue they generated. The hybrid model is proposed, which include five conceptually different machine learning model. A voting mechanism is adapted to increase accuracy. The model has less training and testing computational time than the existing state-of-the-art models. The model is trained on movie data from 1915 to 2015 and predicts success or failure the outcome of movies of 2016 or even 2017. The trained model can make the prediction in the early phase of the film release. The forecast accuracy is high as 85%, which can give basic leadership reference to movie producers and reduces the investment risk (Scikit 2016) (Fig. 12).

A proposed system is an excellent tool for stakeholders in the movie industry. In future research, our review will expand preparing cases to enhance forecast accuracy. In the meantime, more models, for example, feedforward neural network and the convolutional neural network will be connected to our examination. Significant further improvement is potentially available when using more fine-grained classification.

Compliance with ethical standards

Conflicts of interest The author declares that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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